Key Performance Indicators (KPIs)

Project Success Measurement

The following KPIs will be used to evaluate the technical performance, reliability, and usability of the system. The listed targets are **initial acceptance goals**, not guaranteed results. Final targets should be confirmed after the team evaluates the available video data and AWS environment.

5.1 Computer Vision KPIs

KPI	Measurement Method	Initial Target
Person Detection Precision	True positive detections / all positive detections	≥ 90%
Person Detection Recall	True positive detections / all actual people in the labeled test data	≥ 90%
Tracking ID Stability	Evaluate ID switches and IDF1 or HOTA against annotated sequences	Establish a baseline, then improve through tuning
Event Detection Recall	Correctly detected ground-truth events / all ground-truth events	≥ 90% for defined test scenarios
False Alert Rate	Incorrect alerts per hour of processed video	≤ 2 false alerts/hour as an initial goal
Tracking Continuity	Percentage of annotated visible-person frames associated with the correct track	Establish a baseline on the evaluation dataset

5.2 System Performance KPIs

KPI	Measurement Method	Initial Target
Event Processing Latency	Time between the relevant video observation and event creation	≤ 5 seconds for the tested deployment configuration
API Response Time	Measure response latency for common dashboard API requests	95th percentile ≤ 500 ms under the defined test load

Application Availability	Available service time / scheduled service time during the evaluation period	≥ 99% during a defined test window, where practical
Camera Failure Detection	Time between stream failure and system detection	≤ 30 seconds
Evidence Storage Success Rate	Successfully stored evidence items / evidence upload attempts	≥ 99% during testing
Data Persistence	Verify that events and metadata remain retrievable after application restart	100% of successfully committed test records retrievable

5.3 Project Delivery KPIs

KPI	Measurement Method	Target
Milestone Completion	Milestones completed on time / total planned milestones	≥ 90%
Functional Test Pass Rate	Passed functional test cases / total executed test cases	≥ 95% before final delivery
Core Feature Completion	Number of completed mandatory MVP features / total mandatory MVP features	100%
Budget Compliance	Actual AWS expenditure compared with the approved project budget	Remain within the approved budget
Documentation Completion	Required technical and project documents completed	100% of required documents delivered

5.4 KPI Evaluation Methodology

1. Prepare a representative test dataset using authorized video footage.
2. Annotate people, tracking identities where feasible, and expected event timestamps.
3. Separate development and test data to avoid evaluating the model only on examples used during tuning.

4. Run the complete system using a documented hardware and AWS configuration.
5. Calculate detection precision, recall, tracking metrics, false alerts per hour, latency, and reliability.
6. Record the results, compare them against the initial targets, and explain any limitations.

5.5 Definition of Success

The project will be considered successful when the core MVP features work together end to end, the application is deployed and accessible to authorized users, security-related events can be reviewed with supporting evidence, and the evaluation report transparently documents performance, limitations, and cloud costs.

All KPI results must be measured and reported honestly. The system should not be considered accurate or production-ready solely because it successfully processes a demonstration video.